DO YOU KNOW?

Penicillin is known as Queen of Medicine. It was discovered by Sir Alexander Fleming in 1928.

17.3 Bryophytes

Bryophytes are the primitive and simplest group of plants. These are terrestrial and non-vascular cryptogams (They have no vascular tissues like xylem and phloem). Bryophytes live on land and in water. Therefore, they are named as amphibians of plant kingdom. Water is essential to complete their life cycle.

Bryophytes have distinct alternation of generation. Gametophyte generation is dominant and sporophytic generation is small. Sporophytic generation depends on the gametophytic generation. The gametophytic plant can be either thalloid (liverworts) or leafy (mosses). The plant remains fixed to the substratum with the help of root like structure called rhizoid.

Sexual reproduction is oogamous type. They have well developed sex organs like antheridia and archegonia. The male sex organ is antheridium, which produces antherozoid. The female sex organ is archegonium which contains an egg. Antherozoid swims with the help of water and reaches the archegonium. It fertilizes the egg and forms zygote (2n). Zygote is the first cell which develops into sporophytic generation and produces haploid spore (n) by meiosis. Spore is the first cell of the gametophytic generation.

17.3.1 Classification of Bryophytes

Bryophytes are classified into three classes. They are:

1. Hepaticopsida (Liverworts)
2. Anthocerotopsida (Hornworts)
3. Bryopsida (Mosses)

Hepaticopsida (E.g. *Riccia*)

- These are lower forms of bryophytes. They are simple in structure than moss.
- Sporophyte is very simple and short lived.

Anthocerotopsida (E.g. *Anthoceros*)

- Gametophyte is undifferentiated thallus. Rhizoids are unicellular and unbranched.
- Protonemal stage is absent. Sporophyte is differentiated into foot and capsule only.

Bryopsida (E.g. *Funaria*)

- These are higher forms in which the gametophyte is differentiated into stem, leaf and root like parts.
- Protonemal stage is present.
- Sporophytes are differentiated into foot, seta, and capsule.

17.3.2 Economic importance

- Bryophytes prevent soil erosion.
- *Sphagnam* can absorb large amount of water. Hence, it is used by the gardeners in nursery.
- Peat which is a valuable fuel like coal is obtained from *Sphagnum*.



Anthoceros Riccia Funaria

Figure 17.9 Bryophytes

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